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LESSON 1: Library management system

-Identify candidate classes (Books, Readers, Vouchers, Employees, Publishers)

+ Book: main entity, managing book information.

- + DocGia: person who borrows books.

- + PhieuMuon: records borrowing/returning.

- + NhanVien: system manager (librarian, warehouse keeper...).

- + NhaXuatBan: book source information.

 - Properties and behavior (methods) of each class

- + Sach

 - o Attributes: maSach (PK, string), nameSach (string), tacGia (string), namXuatBan (int), manNXB (FK), soLuongTon (int), trangThai (string: "Yes available"/"On loan").

 - o Behavior: addSach(), capNhatSach(), deleteSach(), saveSach(), capNhatTonKho().

- + DocGia

 - o Attributes: maDocGia (PK, string), hoTen (string), daySinh (date), diaChi (string), soDienThoai (string), dayDangKy (date), gioiTinh (boolean).

 - o Behavior: dangKyThe(), capNhatThongTin(), viewLichSuMuon().

- + PhieuMuon

 - o Attributes: maPhieu (PK, string), maDocGia (FK), maSach (FK),

 - FK, you're (date), you're (date), you're (date), TinhTrang (string: "On loan"/"Paid"/"Overdue").

 - o Behavior: createPhieuMuon(), searchSach(), saveTraQuaHan(), crystalPhieuPhat().

- + NhanVien

- o Attributes: maNhanVien (PK, string), hoTen (string), chucVu (string), date of birth (date), soDienThoai (string).

- o Behavior: dangNhap (), quanLyDocGia (), quanLyMuonTra (), quanLySach ().

+ NhaXuatBan

- o Attributes: maNXB (PK, string), nameNXB (string), diaChi (string), email (string).

- o Behavior: addNXB (), capNhatNXB ().

+ Relationships

+Books and Publishers: 1 book – 1 publisher

- +Readers and Borrowing Slips: 1 reader – many borrowing slips
- +Books and loan slips: 1 book - many loan slips
- +Employees and Loan Vouchers: 1 employee – many vouchers

LESSON 2: Online sales management system

- Identify main use cases

Main actors:

- + Customer (Customers / Potential Customers).
- + NguoiQuanLy / Admin (Administrator / Sales staff).

Main use cases (customer registration, ordering, payment, product management):

KhachHang:

- + DangKyTaiKhan (Customer registration).
- + DangNhap (Sign in).
- + View SanPham (Search / View product list).
- + ThemVaoGioHang (Add to cart).
- + DatHang (Order).
- + ThanhToan (Online payment / COD).
- + View DonHang (View order history).

Admin / NguoiQuanLy:

- + QuanLySanPham (Add/Edit/Delete products).
- + QuanLyDonHang (View/Confirm/Cancel order).
- + QuanLyKhachHang (Customer information management).
- + QuanLyThanhToan (Payment processing, refund if necessary).

Extend / Include:

- + ThanhToan includes XacNhanDonHang.
- +DatHang extends ThanhToan (if paying online).

- From use case, find the necessary classes

- +Customers (Customer): from register, view orders.
- +SanPham (Product): product management.
- +GioHang (Cart): temporarily save the product before ordering.
- +Don Hang (Order): from order, payment.
- +ChiTietDonHang (OrderDetail): product details in the order.
- +ThanhToan (Payment): payment processing (may have subclasses such as OnlinePayment, COD).
- +Admin or NguoiQuanLy (inherits from User if there is a user system).

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+ - Analyze relationships between classes (specify inheritance if any) +

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+KhachHang and DonHang: 1 customer – many orders

+DonHang and ChiTietDonHang: 1 order – many order details (Composition)

+ChiTietDonHang and San

Pham: many single details – 1 product

+DonHang and ThanhToan: 1 order – 1 payment

+User (abstract class) ←inherit ←KhachHang and Admin (KhachHang and

Admins are all the same type of User, sharing the same account, password, and full name).

Lesson 3: Convert class diagrams into data tables

1. MonHoc table

No	School name	Data type	Bind			
			PK	FK	Not NULL	Unique
1	maMon	Nvarchar(10)	x		x	x
2	nameMon	Nvarchar(50)			x	
3	soTinChiLT	Int			x	
4	soTinChiTH	Int			x	

2. Science Table

No	School name	Data type	Bind			
			PK	FK	Not NULL	Unique
1	maHoHoc	Nvarchar(10)	x			x
2	tenKhoaHoc	Nvarchar(50)			x	
3	ngayTao	Date			x	